

Quan Trinh

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github.com/quanttrinh 

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SKILLS

Languages: Python, TypeScript, C/C++, Bash, SQL.

Frameworks & Technologies: Django, React, gRPC, Protobuf, Kafka, Docker, CI/CD, Linux.

EDUCATION

University of Waterloo

Waterloo, Ontario

Bachelor of Software Engineering (Honors)

Sep 2023 – Apr 2028

- Relevant courses: Software Engineering Principles, Data Structures and Data Management, Database.

EXPERIENCE

Software Developer | [Carta KYC](#) 

May 2026 – Aug 2026

Carta

Kitchener, Ontario

- Launched Singapore Corporate Secretarial KYC across the Django service, gRPC API, frontend, CLI, and Claude skills, used for **48 evaluations** across **103 entities** in the first month.
- Built name-mismatch detection across entity, portfolio, and government-ID names, screening **15,402 evaluations** and routing **358 flagged evaluations** for compliance review over 11 weeks.
- Restructured CI into a path-filtered pipeline and dynamically sized test workers, reducing the median build time from **15m57s to 8m52s** and credits per run from **3,251 to 1,981**.
- Implemented a document validation pipeline with AI field extraction across the Django service, gRPC API, and Kafka events, enabling form pre-submit warnings and post-submit review.
- Eliminated multiple N+1 query patterns in the Django KYC service, cutting the heaviest document read from **111 ms to 17 ms** and adverse-media loading from **103 ms to 3 ms**.
- Hardened the KYC service Docker image by auditing dependencies and using a non-root, read-only base, reducing vulnerabilities from **163 to 20** and image size from **389 MB to 264 MB**.

Software Developer | [onsemi Studio](#) 

Sep 2025 – Dec 2025

onsemi

Waterloo, Ontario

- Integrated bare-metal CMake project support in onsemi Studio, **15 days** ahead of schedule, enabling internal teams to adopt the extension before the public release deadline.
- Refactored onsemi Studio repo-wide, reducing package size from **11 MB to 5 MB** and resolving **over 300 flagged issues**.
- Created a VS Code embedded project configuration GUI with VS Code Extension API and React, replacing manual JSON editing workflows.

Full-Stack Developer | [onsemi Current Sense Design Tool](#) 

Jan 2025 – Apr 2025

onsemi

Waterloo, Ontario

- Launched a new FastAPI and Next.js-based onsemi Current Sense Design Tool, replacing the existing Excel workflows.
- Created React components and supporting infrastructure for an in-house Electron-based GUI library, enabling the transition from a legacy Qt-based library.
- Built a Protobuf-to-TypeScript code generation CLI using TypeScript to automate the mapping of the device register schema to UI controls, eliminating manual boilerplate and mapping errors.

AI Solution Developer

Jun 2024 – Aug 2024

Microsoft Azure & Artificial Intelligence Fundamentals, WE Accelerate

Waterloo, Ontario

- Designed a solution based on Azure and Azure AI to combat online fraud targeting the elderly.
- Achieved AZ-900 and AI-900 certifications.

PROJECTS

Event Registration Website for Community Lunar Celebration | [GitHub](#)

- Built a full-stack, multilingual event registration web app using SolidJS and Google Apps Script, digitizing the registration process at zero hosting cost.
- Implemented secure registration and check-in process with automatic emails and QR code scanning for contactless check-in, handling **over 200 registrations** on event day.
- Created a custom build/deploy pipeline (Vite plugin, yEnc compression, CLASP integration), reducing payload size by **~20–30%** and enabling flexible deployment.